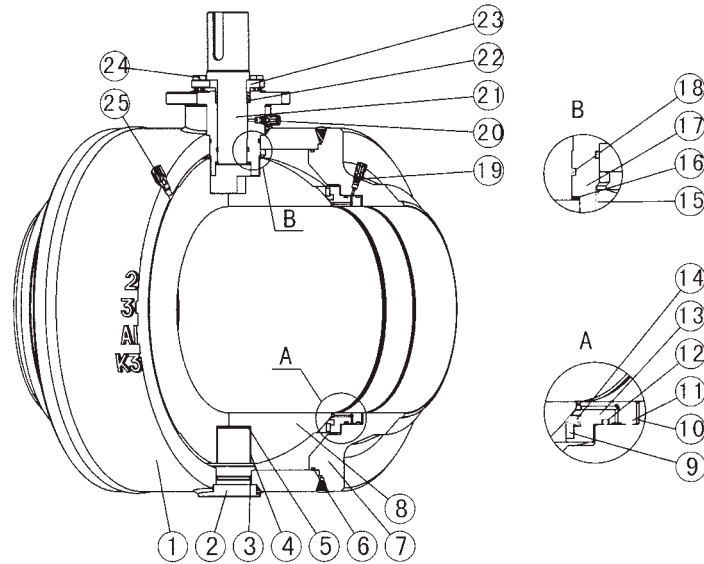


# Welded Structure Ball Valve

Valve Structural Schematic Drawing (Forge Welded Structure)



The Material of Main Parts (Forge Welded)

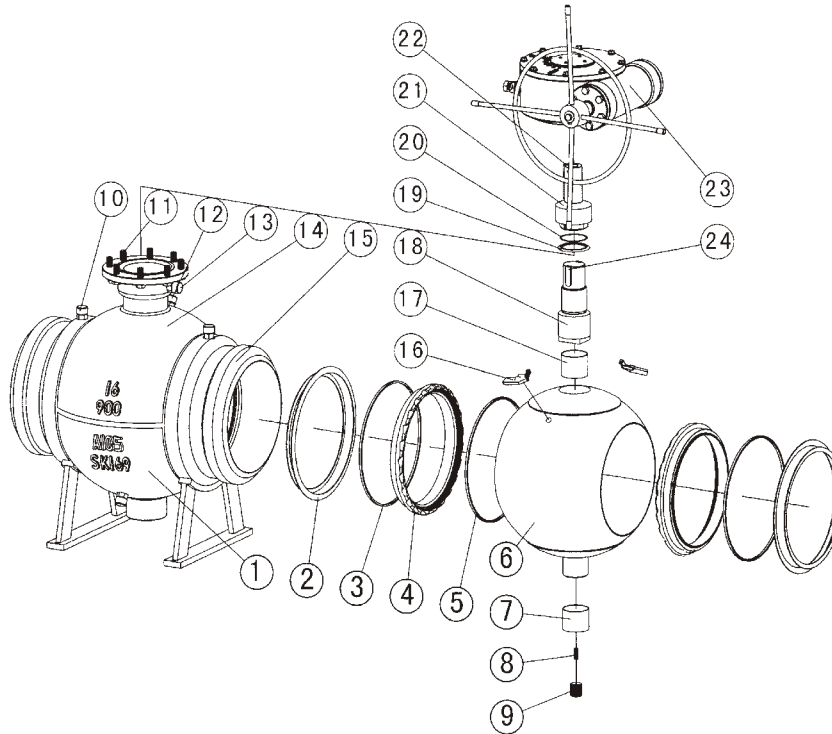
| No. | Parts Name                       | Material                    |           |                       |
|-----|----------------------------------|-----------------------------|-----------|-----------------------|
|     |                                  | Carbon Steel                | SS        | Low Temperature Steel |
| 1   | Body                             | A105                        | A182 F316 | A350 LF3              |
| 2   | Lower Stem Seat                  | A105                        | A182 F316 | A350 LF3              |
| 3   | O-Ring                           | VITON                       |           |                       |
| 4   | Bushing                          | PTFE+CS                     | PTFE+SS   | PTFE+SS               |
| 5   | Flat Bushing                     | PTFE+CS                     | PTFE+SS   | PTFE+SS               |
| 6   | O-Ring                           | VITON                       |           |                       |
| 7   | Bonnet                           | A105                        | A182 F316 | A350 LF3              |
| 8   | Ball                             | A105+ENP                    | A182 F316 | A350 LF3              |
| 9   | Pressboard Ring                  | A105+ENP                    | A182 F316 | A350 LF3              |
| 10  | Spring                           | INCONEL X-750               |           |                       |
| 11  | Fire Safe Ring                   | Soft Graphite               |           |                       |
| 12  | O-Ring                           | VITON                       |           |                       |
| 13  | Seat                             | A105+ENP                    | A182 F316 | A350 LF3              |
| 14  | Sealing Ring                     | PTFE、NYLON、PEEK、PCTFE、MOLON |           |                       |
| 15  | Bushing                          | PTFE+CS                     | PTFE+SS   | PTFE+SS               |
| 16  | Flat Bushing                     | PTFE+CS                     | PTFE+SS   | PTFE+SS               |
| 17  | Upper Stem Seat                  | A105                        | A182 F316 | A350 LF3              |
| 18  | O-Ring                           | VITON                       |           |                       |
| 19  | Seat Ring Grease Injecting Valve | A105+ENP                    | A182 F316 | A350 LF3              |
| 20  | Stem Grease Injecting Valve      | A105+ENP                    | A182 F316 | A350 LF3              |
| 21  | Stem                             | A29 4140+ENP                | A182 F316 | A182 F316             |
| 22  | Packing                          | Soft Graphite               |           |                       |
| 23  | Gland                            | A105+ENP、A29 4140+ENP       | A182 F316 | A350 LF3              |
| 24  | Screw                            | A193 B7                     | A193 B8   | A320 L7               |
| 25  | Discharge Valve                  | A29 1025                    | SS        | SS                    |

Notes: 1. Materials are subject to users' requirements and different industries;

2. Materials of Anti-sulfur valve shall be according to NACE MR0175

# Welded Structure Ball Valve

Valve Structural Schematic Drawing (Plate Welded or Cast Welded)



The Material of Main Parts (Plate Welded and Cast Welded)

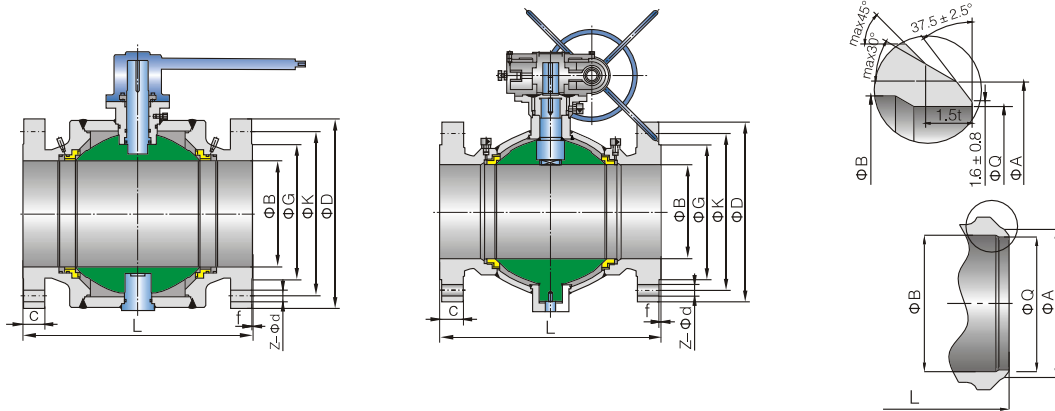
| No. | Parts Name                  | Material              |                     |                       |
|-----|-----------------------------|-----------------------|---------------------|-----------------------|
|     |                             | Carbon Steel          | SS                  | Low Temperature Steel |
| 1   | Lower Body                  | A105/A216 WCB         | A182 F316/A351 CF8M | A350 LF3/A352 LB      |
| 2   | Leaf Spring                 | INCONEL X-750/17-4PH  | Incone1600/17-4PH   | Incone1600/17-4PH     |
| 3   | O-Ring                      | VITON                 |                     |                       |
| 4   | Seat                        | A105+ENP              | A182 F316           | A350 LF3              |
| 5   | Sealing Ring                | PTFE、NYLON、PEEK、PCTFE |                     |                       |
| 6   | Ball                        | A105+ENP/A216 WCB+ENP | A182 F316/A351 CF8M | A350 LF3/A352 LB      |
| 7   | Bushing                     | PTFE+CS               | PTFE+SS             | PTFE+SS               |
| 8   | Spring                      | INCONEL X-750         |                     |                       |
| 9   | Adjusting Screw             | A193 B7               | A182 F316           | A350 LF3              |
| 10  | Seat Grease Injecting Valve | A105+ENP              | A182 F316           | A350 LF3              |
| 11  | Bolt                        | A193 B7               | A193 B8             | A320 L7               |
| 12  | Discharge Valve             | A29 1025              | SS                  | SS                    |
| 13  | Stem Grease Injecting Valve | A29 1025              | SS                  | SS                    |
| 14  | Upper Body                  | A105/A216 WCB         | A182 F316/A351 CF8M | A350 LF3/A352 LB      |
| 15  | Bonnet                      | A105/A216 WCB         | A182 F316/A351 CF8M | A350 LF3/A352 LB      |
| 16  | Pallet Mechanism            | -                     |                     |                       |
| 17  | Upper Collar                | PTFE+CS               | PTFE+SS             | PTFE+SS               |
| 18  | Stem                        | A29 4140+ENP          | A182 F316           | A182 F316             |
| 19  | Gasket                      | PTFE                  |                     |                       |
| 20  | Plane Collar                | PTFE+SS               |                     |                       |
| 21  | Joint Sleeve                | A29 1045              |                     |                       |
| 22  | Key                         | A29 1045              |                     |                       |
| 23  | Worm Gear Device            | -                     |                     |                       |
| 24  | Key                         | A29 1045              |                     |                       |

Notes: 1. Materials are subject to users' requirements and different working conditions;

2. Materials of Anti-sulfur valve shall be according to NACE MR0175;

3. When all internal parts are assembled, get upper body, lower body and bonnet together closely and then weld it after the pressure test is passed.

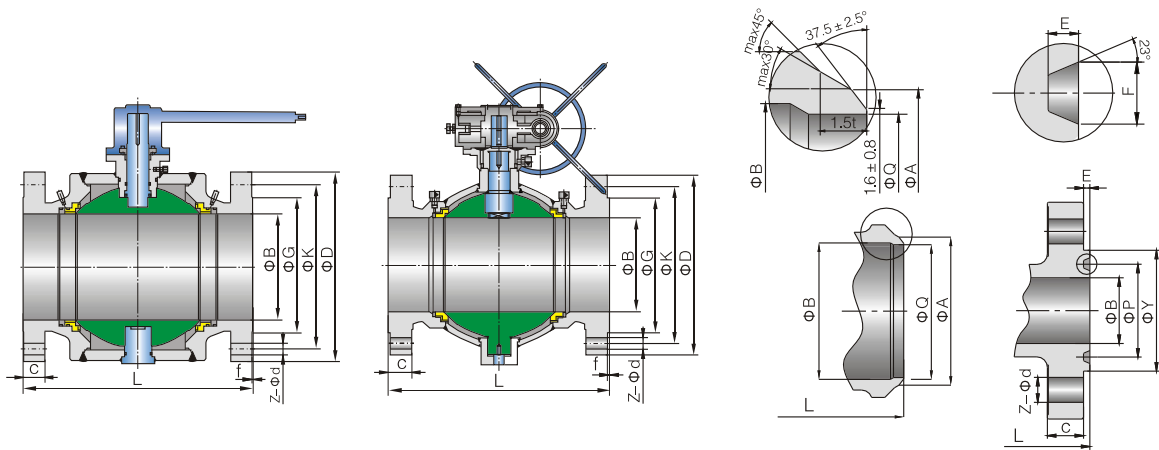
# Full Port Welded Structure Ball Valve



Main Connection Dimension

| Class | Size |      | Dimensions(mm) |      |      |        |        |        |         |         |      |      |                            |
|-------|------|------|----------------|------|------|--------|--------|--------|---------|---------|------|------|----------------------------|
|       | NPS  | DN   | L              |      | B    | D      | G      | f      | K       | Z-Φd    | C    | A    | Q                          |
|       |      |      | RF             | BW   |      |        |        |        |         |         |      |      |                            |
| 150Lb | 2    | 50   | 178            | 216  | 49   | 152    | 91.9   | 1.6    | 120.7   | 4-19.1  | 15.8 | 63   | Asper client's requirement |
|       | 3    | 80   | 203            | 283  | 74   | 191    | 127    | 1.6    | 152.4   | 4-19.1  | 19.1 | 91   |                            |
|       | 4    | 100  | 229            | 305  | 100  | 229    | 157.2  | 1.6    | 190.5   | 8-19.1  | 23.9 | 117  |                            |
|       | 6    | 150  | 394            | 457  | 150  | 279    | 215.9  | 1.6    | 241.3   | 8-22.2  | 25.4 | 172  |                            |
|       | 8    | 200  | 457            | 521  | 201  | 343    | 269.7  | 1.6    | 298.5   | 8-22.2  | 28.4 | 223  |                            |
|       | 10   | 250  | 533            | 559  | 252  | 406    | 323.9  | 1.6    | 362     | 12-25.4 | 30.2 | 278  |                            |
|       | 12   | 300  | 610            | 635  | 303  | 483    | 381    | 1.6    | 431.8   | 12-25.4 | 31.8 | 329  |                            |
|       | 14   | 350  | 686            | 762  | 334  | 533    | 412.8  | 1.6    | 476.2   | 12-28.6 | 35.1 | 362  |                            |
|       | 16   | 400  | 762            | 838  | 385  | 597    | 469.9  | 1.6    | 539.8   | 16-28.6 | 36.6 | 413  |                            |
|       | 18   | 450  | 864            | 914  | 436  | 635    | 533.4  | 1.6    | 577.9   | 16-31.8 | 39.6 | 464  |                            |
|       | 20   | 500  | 914            | 991  | 487  | 699    | 584.2  | 1.6    | 635     | 20-31.8 | 42.9 | 516  |                            |
|       | 24   | 600  | 1067           | 1143 | 589  | 813    | 692.2  | 1.6    | 749.3   | 20-34.9 | 47.8 | 619  |                            |
|       | 26   | 650  | 1143           | 1245 | 633  | 870    | 749.3  | 1.6    | 806.5   | 24-34.9 | 68.3 | 670  |                            |
|       | 28   | 700  | 1245           | 1346 | 684  | 927    | 800.1  | 1.6    | 863.6   | 28-34.9 | 71.4 | 721  |                            |
|       | 30   | 750  | 1295           | 1397 | 735  | 984    | 857.3  | 1.6    | 914.4   | 28-34.9 | 74.7 | 772  |                            |
|       | 32   | 800  | 1372           | 1524 | 779  | 1060   | 914.4  | 1.6    | 977.9   | 28-41.3 | 80.8 | 825  |                            |
|       | 34   | 850  | 1473           | 1626 | 830  | 1111   | 965.2  | 1.6    | 1028.7  | 32-41.3 | 82.6 | 876  |                            |
|       | 36   | 900  | 1524           | 1727 | 874  | 1168   | 1022.4 | 1.6    | 1085.9  | 32-41.3 | 90.4 | 927  |                            |
|       | 40   | 1000 | 1753           | 1956 | 976  | 1289   | 1124   | 1.6    | 1200.2  | 36-41.3 | 90.4 | 1029 |                            |
|       | 42   | 1050 | 1855           | 2083 | 1020 | 1346   | 1193.8 | 1.6    | 1257.3  | 36-41.3 | 96.8 | 1080 |                            |
| 48    | 1200 | 2134 | 2388           | 1166 | 1511 | 1358.9 | 1.6    | 1422.4 | 44-41.3 | 108     | 1232 |      |                            |

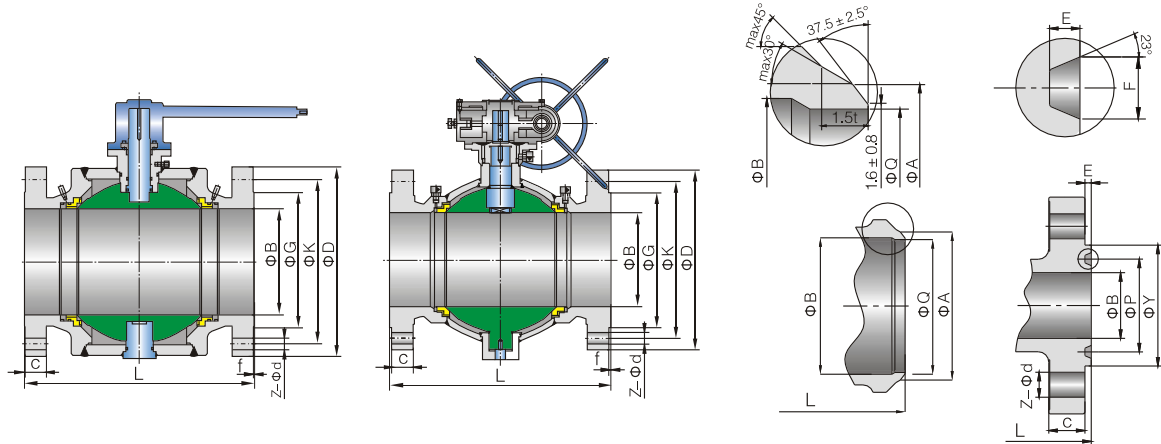
# Full Port Welded Structure Ball Valve



Main Connection Dimension

| Class | Size  |     | Dimensions(mm) |      |      |     |      |        |     |        |         |       |        |         |       |       |     |                            |
|-------|-------|-----|----------------|------|------|-----|------|--------|-----|--------|---------|-------|--------|---------|-------|-------|-----|----------------------------|
|       | NPS   | DN  | L              |      |      | B   | D    | G      | f   | K      | Z-Φd    | C     | Groove |         |       |       | A   | Q                          |
|       |       |     | RF             | RTJ  | BW   |     |      |        |     |        |         |       | Y      | P       | E     | F     |     |                            |
| 300Lb | 2     | 50  | 216            | 232  | 216  | 49  | 165  | 91.9   | 1.6 | 127    | 8-19.1  | 22.4  | 108    | 82.55   | 7.92  | 11.91 | 63  | Asper client's requirement |
|       | 2 1/2 | 65  | 241            | 257  | 241  | 62  | 191  | 104.6  | 1.6 | 149.2  | 8-22.2  | 25.4  | 127    | 101.6   | 7.92  | 11.91 | 75  |                            |
|       | 3     | 80  | 283            | 298  | 283  | 74  | 210  | 127    | 1.6 | 168.3  | 8-22.2  | 28.4  | 146    | 123.82  | 7.92  | 11.91 | 91  |                            |
|       | 4     | 100 | 305            | 321  | 305  | 100 | 254  | 157.2  | 1.6 | 200.2  | 8-22.2  | 31.8  | 175    | 149.22  | 7.92  | 11.91 | 117 |                            |
|       | 6     | 150 | 403            | 419  | 457  | 150 | 318  | 215.9  | 1.6 | 269.8  | 12-22.2 | 36.6  | 241    | 211.12  | 7.92  | 11.91 | 172 |                            |
|       | 8     | 200 | 502            | 518  | 521  | 201 | 381  | 269.7  | 1.6 | 330.2  | 12-25.4 | 41.1  | 302    | 269.88  | 7.92  | 11.91 | 223 |                            |
|       | 10    | 250 | 568            | 584  | 559  | 252 | 445  | 323.9  | 1.6 | 387.4  | 16-28.6 | 47.8  | 356    | 323.85  | 7.92  | 11.91 | 278 |                            |
|       | 12    | 300 | 648            | 664  | 635  | 303 | 521  | 381    | 1.6 | 450.9  | 16-31.8 | 50.8  | 413    | 381     | 7.92  | 11.91 | 329 |                            |
|       | 14    | 350 | 762            | 787  | 762  | 334 | 584  | 412.8  | 1.6 | 514.4  | 20-31.8 | 53.8  | 457    | 419.1   | 7.92  | 11.91 | 362 |                            |
|       | 16    | 400 | 838            | 854  | 838  | 385 | 648  | 469.9  | 1.6 | 571.5  | 20-34.9 | 57.2  | 508    | 469.9   | 7.92  | 11.91 | 413 |                            |
|       | 18    | 450 | 914            | 930  | 914  | 436 | 711  | 533.4  | 1.6 | 628.7  | 24-34.9 | 60.5  | 575    | 533.4   | 7.92  | 11.91 | 464 |                            |
|       | 20    | 500 | 991            | 1010 | 991  | 487 | 775  | 584.2  | 1.6 | 685.8  | 24-34.9 | 63.5  | 635    | 584.2   | 9.52  | 13.49 | 516 |                            |
|       | 24    | 600 | 1143           | 1165 | 1143 | 589 | 914  | 692.2  | 1.6 | 812.8  | 24-41.3 | 69.9  | 749    | 692.15  | 11.13 | 16.66 | 619 |                            |
|       | 26    | 650 | 1245           | 1270 | 1245 | 633 | 972  | 749.3  | 1.6 | 876.3  | 28-44.5 | 79.2  | 809.8  | 749.3   | 12.7  | 19.84 | 670 |                            |
|       | 28    | 700 | 1346           | 1372 | 1346 | 684 | 1035 | 800.1  | 1.6 | 939.8  | 28-44.5 | 85.9  | 860.6  | 800.1   | 12.7  | 19.84 | 721 |                            |
|       | 30    | 750 | 1397           | 1422 | 1397 | 735 | 1092 | 857.3  | 1.6 | 997    | 28-47.6 | 91.9  | 917.4  | 857.25  | 12.7  | 19.84 | 772 |                            |
|       | 32    | 800 | 1524           | 1533 | 1524 | 779 | 1149 | 914.4  | 1.6 | 1054.1 | 28-50.8 | 98.6  | 984.3  | 914.4   | 14.27 | 23.01 | 825 |                            |
|       | 34    | 850 | 1626           | 1654 | 1626 | 830 | 1207 | 965.2  | 1.6 | 1104.9 | 28-50.8 | 101.6 | 1035.1 | 965.2   | 14.27 | 23.01 | 876 |                            |
|       | 36    | 900 | 1727           | 1756 | 1727 | 874 | 1270 | 1022.4 | 1.6 | 1168.4 | 32-54   | 104.6 | 1092.2 | 1022.35 | 14.27 | 23.01 | 927 |                            |

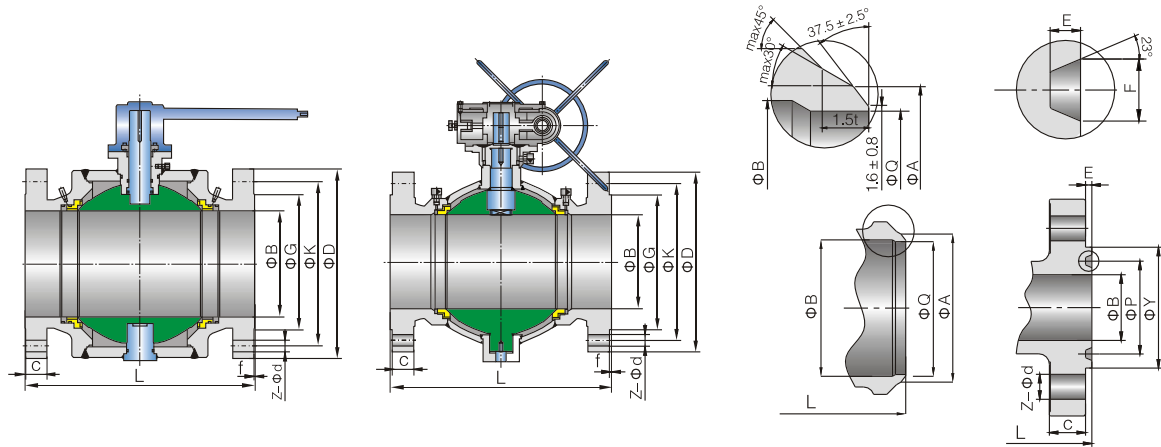
# Full Port Welded Structure Ball Valve



Main Connection Dimension

| Class | Size  |     | Dimensions(mm) |      |      |     |      |        |     |        |         |       |        |         |       |       |     |                            |
|-------|-------|-----|----------------|------|------|-----|------|--------|-----|--------|---------|-------|--------|---------|-------|-------|-----|----------------------------|
|       | NPS   | DN  | L              |      |      | B   | D    | G      | f   | K      | Z-Φd    | C     | Groove |         |       |       | A   | Q                          |
|       |       |     | RF             | RTJ  | BW   |     |      |        |     |        |         |       | Y      | P       | E     | F     |     |                            |
| 600Lb | 2     | 50  | 292            | 295  | 292  | 49  | 165  | 91.9   | 6.4 | 127    | 8-19.1  | 25.4  | 108    | 82.55   | 7.92  | 11.91 | 63  | Asper client's requirement |
|       | 2 1/2 | 65  | 330            | 333  | 330  | 62  | 191  | 104.6  | 6.4 | 149.2  | 8-22.2  | 28.4  | 127    | 101.6   | 7.92  | 11.91 | 75  |                            |
|       | 3     | 80  | 356            | 359  | 356  | 74  | 210  | 127    | 6.4 | 168.3  | 8-22.2  | 31.8  | 146    | 123.82  | 7.92  | 11.91 | 91  |                            |
|       | 4     | 100 | 432            | 435  | 432  | 100 | 273  | 157.2  | 6.4 | 215.9  | 8-25.4  | 38.1  | 175    | 149.22  | 7.92  | 11.91 | 117 |                            |
|       | 6     | 150 | 559            | 562  | 559  | 150 | 356  | 215.9  | 6.4 | 292.1  | 12-28.6 | 47.8  | 241    | 211.12  | 7.92  | 11.91 | 172 |                            |
|       | 8     | 200 | 660            | 664  | 660  | 201 | 419  | 269.7  | 6.4 | 349.2  | 12-31.8 | 55.6  | 302    | 269.88  | 7.92  | 11.91 | 223 |                            |
|       | 10    | 250 | 787            | 791  | 787  | 252 | 508  | 323.9  | 6.4 | 431.8  | 16-34.9 | 63.5  | 356    | 323.85  | 7.92  | 11.91 | 278 |                            |
|       | 12    | 300 | 838            | 841  | 838  | 303 | 559  | 381    | 6.4 | 489    | 20-34.9 | 66.5  | 413    | 381     | 7.92  | 11.91 | 329 |                            |
|       | 14    | 350 | 889            | 892  | 889  | 334 | 603  | 412.8  | 6.4 | 527.1  | 20-38.1 | 69.9  | 457    | 419.1   | 7.92  | 11.91 | 362 |                            |
|       | 16    | 400 | 991            | 994  | 991  | 385 | 686  | 469.9  | 6.4 | 603.2  | 20-41.3 | 76.2  | 508    | 469.9   | 7.92  | 11.91 | 413 |                            |
|       | 18    | 450 | 1092           | 1095 | 1092 | 436 | 743  | 533.4  | 6.4 | 654.1  | 20-44.5 | 82.6  | 575    | 533.4   | 7.92  | 11.91 | 464 |                            |
|       | 20    | 500 | 1194           | 1200 | 1194 | 487 | 813  | 584.2  | 6.4 | 723.9  | 24-44.5 | 88.9  | 635    | 584.2   | 9.52  | 13.49 | 516 |                            |
|       | 24    | 600 | 1397           | 1407 | 1397 | 589 | 940  | 692.2  | 6.4 | 838.2  | 24-50.8 | 101.6 | 749    | 692.15  | 11.13 | 16.66 | 619 |                            |
|       | 26    | 650 | 1448           | 1461 | 1448 | 633 | 1016 | 749.3  | 6.4 | 914.4  | 28-50.8 | 108   | 809.8  | 749.3   | 12.7  | 19.84 | 670 |                            |
|       | 28    | 700 | 1549           | 1562 | 1549 | 684 | 1073 | 800.1  | 6.4 | 965.2  | 28-54   | 111.3 | 860.6  | 800.1   | 12.7  | 19.84 | 721 |                            |
|       | 30    | 750 | 1651           | 1664 | 1651 | 735 | 1130 | 857.3  | 6.4 | 1022.4 | 28-54   | 114.3 | 917.4  | 857.25  | 12.7  | 19.84 | 772 |                            |
|       | 32    | 800 | 1778           | 1794 | 1778 | 779 | 1194 | 914.4  | 6.4 | 1079.5 | 28-60.3 | 117.3 | 984.3  | 914.4   | 14.27 | 23.01 | 825 |                            |
|       | 34    | 850 | 1930           | 1946 | 1930 | 830 | 1245 | 965.2  | 6.4 | 1130.3 | 28-60.3 | 120.7 | 1035.1 | 965.2   | 14.27 | 23.01 | 876 |                            |
|       | 36    | 900 | 2083           | 2099 | 2083 | 874 | 1314 | 1022.4 | 6.4 | 1193.8 | 28-66.7 | 124   | 1092.2 | 1022.35 | 14.27 | 23.01 | 927 |                            |

# Full Port Welded Structure Ball Valve



Main Connection Dimension

| Class | Size  |     | Dimensions(mm) |      |      |     |      |       |     |       |         |       |        |        |       |       |     |                            |
|-------|-------|-----|----------------|------|------|-----|------|-------|-----|-------|---------|-------|--------|--------|-------|-------|-----|----------------------------|
|       | NPS   | DN  | L              |      |      | B   | D    | G     | f   | K     | Z-Φd    | C     | Groove |        |       |       | A   | Q                          |
|       |       |     | RF             | RTJ  | BW   |     |      |       |     |       |         |       | Y      | P      | E     | F     |     |                            |
| 900Lb | 2     | 50  | 368            | 371  | 368  | 49  | 216  | 91.9  | 6.4 | 165.1 | 8-25.4  | 38.1  | 124    | 95.25  | 7.92  | 11.91 | 63  | Asper client's requirement |
|       | 2 1/2 | 65  | 419            | 422  | 419  | 62  | 244  | 104.6 | 6.4 | 190.5 | 8-28.6  | 41.1  | 137    | 107.95 | 7.92  | 11.91 | 75  |                            |
|       | 3     | 80  | 381            | 384  | 381  | 74  | 241  | 127   | 6.4 | 190.5 | 8-25.4  | 38.1  | 156    | 123.82 | 7.92  | 11.91 | 91  |                            |
|       | 4     | 100 | 457            | 460  | 457  | 100 | 292  | 157.2 | 6.4 | 235   | 8-31.8  | 44.5  | 181    | 149.22 | 7.92  | 11.91 | 117 |                            |
|       | 6     | 150 | 610            | 613  | 610  | 150 | 381  | 215.9 | 6.4 | 317.5 | 12-31.8 | 55.6  | 241    | 211.12 | 7.92  | 11.91 | 172 |                            |
|       | 8     | 200 | 737            | 740  | 737  | 201 | 470  | 269.7 | 6.4 | 393.7 | 12-38.1 | 63.5  | 308    | 269.88 | 7.92  | 11.91 | 223 |                            |
|       | 10    | 250 | 838            | 841  | 838  | 252 | 546  | 323.9 | 6.4 | 469.9 | 16-38.1 | 69.9  | 362    | 323.85 | 7.92  | 11.91 | 278 |                            |
|       | 12    | 300 | 965            | 968  | 965  | 303 | 610  | 381   | 6.4 | 533.4 | 20-38.1 | 79.2  | 419    | 381    | 7.92  | 11.91 | 329 |                            |
|       | 14    | 350 | 1029           | 1038 | 1029 | 322 | 641  | 412.8 | 6.4 | 558.8 | 20-41.3 | 85.9  | 467    | 419.1  | 11.13 | 16.66 | 362 |                            |
|       | 16    | 400 | 1130           | 1140 | 1130 | 373 | 705  | 469.9 | 6.4 | 616   | 20-44.5 | 88.9  | 524    | 469.9  | 11.13 | 16.66 | 413 |                            |
|       | 18    | 450 | 1219           | 1232 | 1219 | 423 | 787  | 533.4 | 6.4 | 685.8 | 20-50.8 | 101.6 | 594    | 533.4  | 12.7  | 19.84 | 464 |                            |
|       | 20    | 500 | 1321           | 1334 | 1321 | 471 | 857  | 584.2 | 6.4 | 749.3 | 20-54   | 108   | 648    | 584.2  | 12.7  | 19.84 | 516 |                            |
|       | 24    | 600 | 1549           | 1568 | 1549 | 570 | 1041 | 692.2 | 6.4 | 901.7 | 20-66.7 | 139.7 | 772    | 692.15 | 15.88 | 26.97 | 619 |                            |

| Class  | Size  |     | Dimensions(mm) |      |      |     |      |       |     |       |         |       |        |        |       |       |     |                            |
|--------|-------|-----|----------------|------|------|-----|------|-------|-----|-------|---------|-------|--------|--------|-------|-------|-----|----------------------------|
|        | NPS   | DN  | L              |      |      | B   | D    | G     | f   | K     | Z-Φd    | C     | Groove |        |       |       | A   | Q                          |
|        |       |     | RF             | RTJ  | BW   |     |      |       |     |       |         |       | Y      | P      | E     | F     |     |                            |
| 1500Lb | 2     | 50  | 368            | 371  | 368  | 49  | 216  | 91.9  | 6.4 | 165.1 | 8-25.4  | 38.1  | 124    | 95.25  | 7.92  | 11.91 | 63  | Asper client's requirement |
|        | 2 1/2 | 65  | 419            | 422  | 419  | 62  | 244  | 104.6 | 6.4 | 190.5 | 8-28.6  | 41.1  | 137    | 107.95 | 7.92  | 11.91 | 75  |                            |
|        | 3     | 80  | 470            | 473  | 470  | 74  | 267  | 127   | 6.4 | 203.2 | 8-31.8  | 47.8  | 168    | 136.52 | 7.92  | 11.91 | 91  |                            |
|        | 4     | 100 | 546            | 549  | 546  | 100 | 311  | 157.2 | 6.4 | 241.3 | 8-34.9  | 53.8  | 194    | 161.92 | 7.92  | 11.91 | 117 |                            |
|        | 6     | 150 | 705            | 711  | 705  | 144 | 394  | 215.9 | 6.4 | 317.5 | 12-38.1 | 82.6  | 248    | 211.12 | 9.52  | 13.49 | 172 |                            |
|        | 8     | 200 | 832            | 841  | 832  | 192 | 483  | 269.7 | 6.4 | 393.7 | 12-44.5 | 91.9  | 318    | 269.88 | 11.13 | 16.66 | 223 |                            |
|        | 10    | 250 | 991            | 1000 | 991  | 239 | 584  | 323.9 | 6.4 | 482.6 | 12-50.8 | 108   | 371    | 323.85 | 11.13 | 16.66 | 278 |                            |
|        | 12    | 300 | 1130           | 1146 | 1130 | 287 | 673  | 381   | 6.4 | 571.5 | 16-54   | 124   | 438    | 381    | 14.27 | 23.01 | 329 |                            |
|        | 14    | 350 | 1257           | 1276 | 1257 | 315 | 749  | 412.8 | 6.4 | 635   | 16-60.3 | 133.4 | 489    | 419.1  | 15.88 | 26.97 | 362 |                            |
|        | 16    | 400 | 1384           | 1407 | 1384 | 360 | 826  | 469.9 | 6.4 | 704.9 | 16-66.7 | 146.1 | 546    | 469.9  | 17.48 | 30.18 | 413 |                            |
|        | 18    | 450 | 1537           | 1559 | 1537 | 406 | 914  | 533.4 | 6.4 | 774.7 | 16-73   | 162.1 | 613    | 533.4  | 17.48 | 30.18 | 464 |                            |
|        | 20    | 500 | 1664           | 1686 | 1664 | 454 | 984  | 584.2 | 6.4 | 831.9 | 16-79.4 | 177.8 | 673    | 584.2  | 17.48 | 33.32 | 516 |                            |
|        | 24    | 600 | 1943           | 1972 | 1943 | 546 | 1168 | 692.2 | 6.4 | 990.6 | 16-92   | 203.2 | 794    | 692.15 | 20.62 | 36.53 | 619 |                            |
| 2500Lb | 2     | 50  | 451            | 454  | 451  | 42  | 235  | 91.9  | 6.4 | 171.5 | 8-28.6  | 50.8  | 133    | 101.6  | 7.92  | 11.91 | 63  | Asper client's requirement |
|        | 2 1/2 | 65  | 508            | 514  | 508  | 52  | 267  | 104.6 | 6.4 | 196.9 | 8-31.8  | 57.2  | 149    | 111.12 | 9.52  | 13.49 | 75  |                            |
|        | 3     | 80  | 578            | 584  | 578  | 62  | 305  | 127   | 6.4 | 228.6 | 8-34.9  | 66.5  | 168    | 127    | 9.52  | 13.49 | 91  |                            |
|        | 4     | 100 | 673            | 683  | 673  | 87  | 356  | 157.2 | 6.4 | 273.1 | 8-41.3  | 76.2  | 203    | 157.18 | 11.13 | 16.66 | 117 |                            |
|        | 6     | 150 | 914            | 927  | 914  | 131 | 483  | 215.9 | 6.4 | 368.3 | 8-54    | 108   | 279    | 228.6  | 12.7  | 19.84 | 172 |                            |
|        | 8     | 200 | 1022           | 1038 | 1022 | 179 | 552  | 269.7 | 6.4 | 438.2 | 12-54   | 127   | 340    | 279.4  | 14.27 | 23.01 | 223 |                            |
|        | 10    | 250 | 1270           | 1292 | 1270 | 223 | 673  | 323.9 | 6.4 | 539.7 | 12-66.7 | 165.1 | 425    | 342.9  | 17.48 | 30.18 | 278 |                            |
|        | 12    | 300 | 1422           | 1445 | 1422 | 265 | 762  | 381   | 6.4 | 619.2 | 12-73   | 184.2 | 495    | 406.4  | 17.48 | 33.32 | 329 |                            |

Asper client's requirement